Assignment 1

Question 1. What are the different form factors used in servers? Mention their purpose.

Ans. A server form factor refers to the physical size, shape, and layout of a server hardware. It determines how the server is installed and how much space it occupies.

1. Tower Server

Purpose:

- Looks like a normal desktop computer.
- Used in small businesses or offices.

Advantages:

- Low cost
- Easy to upgrade

2. Rack Server

Purpose:

- Designed to be mounted inside a server rack cabinet.
- Used in data centers and large organizations.

Advantages:

- Saves space
- Easy to manage multiple servers

3. Blade Server

Purpose:

- Very thin servers called blades inserted into a chassis.
- Used in large data centers and cloud environments.

Advantages:

- Very compact
- Less power consumption

4. Micro Server

Purpose:

- Small and energy-efficient server.
- Used for light workloads, hosting, and web services.

Advantages:

- Low power usage
- Cost effective

Question 2. What is the purpose of

Ans. **(i) Netstat**

Netstat stands for Network Statistics.

Purpose:

- Displays network connections
- Shows open ports

Example Use:

- Checking which programs are using network connections.

(ii) MSConfig

MSConfig stands for Microsoft System Configuration Utility.

Purpose:

- Used to configure how Windows starts.
- Manage startup programs.

Example Use:

- Disable unnecessary startup applications to speed up boot time.

(iii) DxDiag

DxDiag stands for DirectX Diagnostic Tool.

Purpose:

- Provides information about graphics, sound, and DirectX components.
- Helps diagnose multimedia and gaming issues.

Example Use:

- Checking GPU details for games.

(iv) Ipconfig

Ipconfig stands for Internet Protocol Configuration.

Purpose:

- Displays IP address of the computer.
- Shows subnet mask and default gateway.

Example Use:

ipconfig /all

This command shows full network configuration.

(v) BCDedit

BCDEdit stands for Boot Configuration Data Editor.

Purpose:

- Used to manage boot configuration data.
- Helps configure boot loader settings.

Example Use:

- Repairing boot entries in Windows.

(vi) Sysinfo32

Sysinfo32 or System Information Tool displays detailed information about the computer.

Purpose:

- Shows hardware information
- Displays operating system details

Example Use:

- Checking system specifications.

Question 3. Describe the booting process of Windows Operating System.

Ans. Booting is the process of starting a computer and loading the operating system into memory.

Steps of Windows Booting Process

1. Power On

- When the computer is turned on, power is supplied to hardware components.
- The CPU starts executing instructions stored in ROM.

2. BIOS / UEFI Execution

The BIOS (Basic Input Output System) or UEFI firmware starts.

Functions:

- Checks hardware components
- Performs POST (Power On Self Test)

3. POST (Power On Self Test)

The system checks hardware like:

- RAM
- Keyboard
- Hard disk
- CPU

If any hardware fails, the system shows an error.

4. Boot Loader Execution

After POST, BIOS/UEFI looks for a bootable device such as:

- Hard drive
- SSD
- USB drive
- DVD

It loads the Windows Boot Manager.

5. Windows Boot Manager

The boot manager reads the Boot Configuration Data (BCD) and selects which operating system to load.

6. Windows Kernel Loading

The system loads the Windows kernel (ntoskrnl.exe) into memory.

The kernel:

- Manages hardware
- Controls system resources
- Starts system services

7. Driver Initialization

Device drivers for hardware like:

- keyboard
- mouse
- disk
- display

8. Login Screen

Finally, Windows loads the user interface and shows the login screen, where the user enters username and password.